2.3.5 External systems

External systems provide context that is important when building a digital representation of a physical system. In many industries, systems external to your digital twin (but internal to your organization), such as *enterprise resource planning (ERP)* and SCADA systems, will contain information about your equipment, its maintenance history, performance and so on. Some external systems may offer *application programming interface (API)* access, making integration technically straightforward via RESTful APIs, GraphQL endpoints, and webhook subscriptions that provide real-time or near-real-time data feeds, while others may need significant effort to access the data they hold.

In my home digital twin, information that can be used to achieve the objectives is available from several external systems. I can access weather forecast data generated from the European Centre for Medium-Range Weather Forecasts (ECMWF) models that we saw in chapter 1, which can be used to predict rainfall and automatically adjust the irrigation system. My electricity provider provides daily electricity usage data via their website that can be used to inform the system more regularly than a quarterly bill. This contextual information is what I will use to move from reactive problem solving towards predictive optimization.

TRY IT OUT: GET FORECAST RAINFALL FROM THE ECMWF

A subset of ECMWF real-time forecast data is made available to the public and can be accessed for free up to ten times a day from the Open-Meteo service (https://open-meteo.com). The following URL, when pasted into a browser, will return the rainfall forecast over the next 7 days from the ECMWF's Integrated Forecasting System (IFS) for the city of Perth.

```
https://api.open-meteo.com/v1/forecast?latitude=-31.9522&longitude=1
15.
➥8614&hourly=rain&models=ecmwf_ifs025&forecast_days=7
```

I will integrate a daily call to this API by my digital twin to provide information that I can use to make dynamic decisions about whether to irrigate the garden or not.